16 EXCEPTION HANDLING • PRACTICE QUESTIONS

conceptual questions

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

Conceptual Questions — Module 16: Exception Handling

Q1. Differentiate between compile-time errors, synchronous runtime exceptions, and hardware traps/Undefined Behavior in C++.

Q2. Explain **Stack Unwinding** in detail. In what order are local stack objects destroyed when an exception is thrown inside a `try` block? What happens if an object's constructor throws an exception halfway through its body?

Q3. What is the difference between rethrowing an exception using `throw;` versus `throw e;` inside a `catch (const BaseException& e)` block?

Q4. Define and compare the **Basic**, **Strong**, and **No-Throw (`noexcept`)** exception safety guarantees with an example of each. Why must destructors in C++ never throw exceptions?